Bonds and yield curves

Pricing cash flows, measuring risk and reading market expectations

Today's route

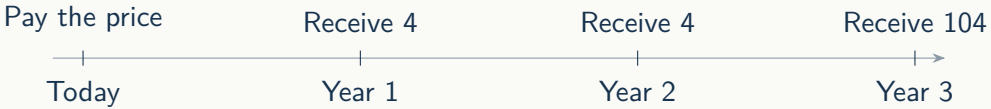
Section	Question
Cash flows and prices	What exactly are we buying?
Bills and auctions	How does a new security get priced?
Curves and forwards	What can we lock in today?
Expectations and term premia	Which part of a yield is a forecast?
Duration and convexity	How much do prices move?
Hedging and market cases	What position expresses the view?

LECTURE 3

Cash flows and prices

What is promised, what is priced, and what can we earn?

A bond is a dated payment contract



$$F = 100, \qquad c = 4\%, \qquad C = cF = 4.$$

Prices and cash-flow symbols

Symbol	Meaning
F	Face value, normally 100 in our examples
$p_t(T)$	Bond price at time t, with T years remaining
$C = cF$	Coupon payment from coupon rate c
$d_t(T)$	Price today of one certain unit paid in T years

Rate symbols and horizons

Symbol	Meaning
$i_t(T)$	Nominal spot rate for a T-year zero
$f_t(T_1, T_2)$	Rate fixed today for a future lending interval
i^{YTM}	One yield matching a coupon bond's price
$\bar{i}_t$	Central bank's chosen short-rate target

r still denotes a real rate. Pricing examples use nominal rates in decimals.

A zero-coupon bond

$$p_t(T) = \frac{F}{[1 + i_t(T)]^T}.$$

$$F = 100, \quad T = 2, \quad i_t(2) = 5\% \quad \Rightarrow \quad p_t(2) = 90.70.$$

A discount factor prices one future unit

$$d_t(T) = \frac{p_t^{\text{zero}}(T)}{F} = \frac{1}{[1 + i_t(T)]^T}.$$

$$p_t = \sum_{k=1}^{n} d_t(k)\, CF_k.$$

CF_k: the cash payment at date k. Match dates, currency and credit risk.

Replication gives the pricing intuition

Buy the same cash flows using zero-coupon bonds.

A matching payoff schedule has a matching price.

Same currency, payment dates and credit quality. Borrowing, shorting and trading must be feasible.

A coupon bond is a basket of zeros

Date	Cash flow	Replication
1	4	4 units of a date-1 zero
2	4	4 units of a date-2 zero
3	104	104 units of a date-3 zero

$$p_t = 4d_t(1) + 4d_t(2) + 104d_t(3).$$

Worked price with a flat curve

$$p_t = \frac{4}{1.05} + \frac{4}{1.05^2} + \frac{104}{1.05^3} = 97.28.$$

A 4% coupon trades below face value when the required yield is 5%.

Annual coupons. Flat 5% spot curve. Valuation on a coupon date.

Market price determines yield to maturity

$$p_t = \sum_{k=1}^{n} \frac{CF_k}{(1 + i^{\mathrm{YTM}})^k}.$$

Given the price and promised payments, solve for the single rate.

YTM summarizes the bond price. Each dated spot rate can still be different.

Coupon, yield and return

Object	What does it measure?
Coupon rate c	Contractual annual payment per unit face
Yield i^{YTM}	Discount rate matching today's price
Holding return H	Cash received plus price change, over your horizon

$$H_{t,t+1} = \frac{C_{t+1} + p_{t+1} - p_t}{p_t}.$$

A 5% yield can produce a one-year loss

Ten-year zero, face 100	Price
Buy at 5% today	61.39
Sell one year later, remaining nine-year yield 6%	59.19

$$H = \frac{59.1898 - 61.3913}{61.3913} = -3.59\%.$$

What holding to maturity locks in

A default-free zero locks its nominal payment and annualized return.

Coupon-bond wealth also depends on coupon reinvestment.

Selling early introduces price risk. Inflation and foreign-currency returns add other risks.

Risk-free depends on the investor's horizon

Need	Relevant risk
100 in ten years	A matched ten-year zero fixes that nominal payment
Spend after one year	A long zero has uncertain resale value
Roll bills for ten years	Future bill reinvestment rates are unknown

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Bills and auctions

How does a new Treasury security get priced?

Treasury bills are short zero-coupon claims

No periodic coupon. The investor receives face value at maturity.

Regular original terms	Minimum purchase
4, 6, 8, 13, 17, 26 and 52 weeks	100 dollars

TreasuryDirect rules checked 16 October 2026. Cash-management bills can have other terms.

How the auction clears

Treasury announces the security, offering size and dates.

Eligible noncompetitive bids receive their requested amounts.

Competitive bids enter from low rates to high rates until the offering is allocated.

Two ways to submit a bid

Noncompetitive	Competitive
Specify the face amount.	Specify face amount and required rate.
Accept the auction's clearing rate.	A bid can be accepted, partly filled or rejected.

Successful bidders pay the uniform auction price.

An invented bill auction

Bidder	Requested face (millions)	Discount-rate bid
A	30	3.90%
B	30	4.00%
C	30	4.10%
D	20	4.20%

Offering: 100 million. Noncompetitive bids: 20 million. Competitive allocation remaining: 80 million.

The cutoff bidder receives a partial fill

Bidder	Accepted face (millions)	Outcome
A	30	Full
B	30	Full
C	20 of 30	Two-thirds
D	0	Rejected

Stop-out discount rate $= 4.10\%$.

One price for every successful bidder

$$p = F\left(1 - i_d \frac{N}{360}\right).$$

$$F = 100, \quad i_d = 4.10\%, \quad N = 28 \quad \Rightarrow \quad p = 99.681111.$$

i_d: bank-discount quotation. N: calendar days from issue to maturity.

Actual four-week auction: 15 October 2026

Reported item	Official result
Public offering	110 billion dollars
High discount rate	4.000%
Price per 100 face	99.688889
Reported bid-to-cover	2.56

Issue: 20 October. Maturity: 17 November. CUSIP 912797VX2. Source: TreasuryDirect.

The same bill has different rate quotations

$$H = \frac{F - p}{p} = 0.3121\% \quad \text{over 28 days.}$$

$$i_d = \frac{F - p}{F}\frac{360}{28} = 4.000\%.$$

$$i_{\text{inv}} = \frac{F - p}{p}\frac{365}{28} = 4.068\%.$$

Same cash flow. Different denominators and annualization rules.

Coupon bonds also have clean and dirty prices

$$p_{\text{dirty}} = p_{\text{clean}} + \text{accrued interest.}$$

$$99.20 + 0.80 = 100.00 \quad \text{per 100 face.}$$

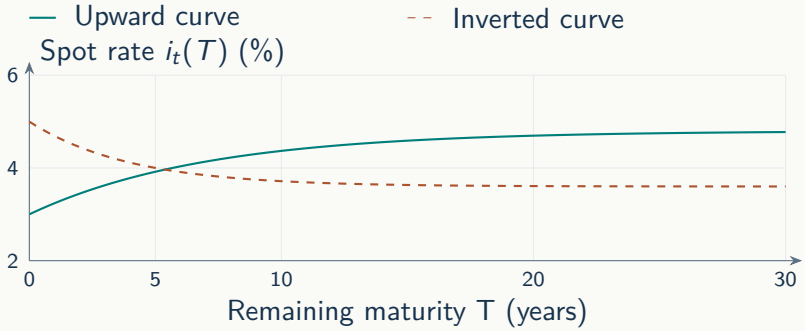
The dirty price is the cash exchanged at settlement.

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Curves and forwards

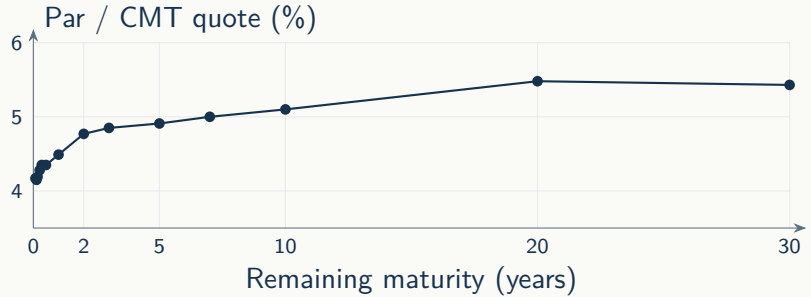
What do prices at different maturities let us lock in?

A yield curve is one date, many maturities



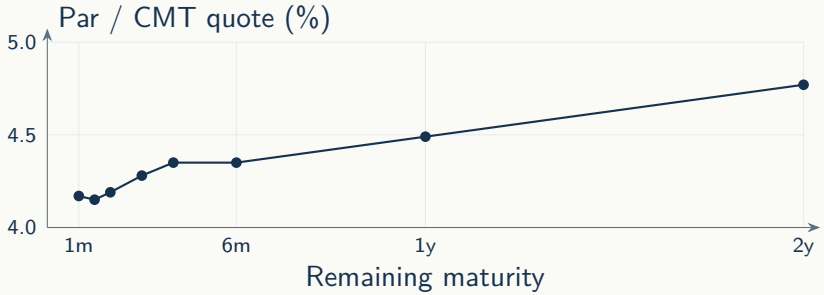
Hypothetical shapes. The horizontal axis is maturity, not the future calendar path of policy.

US Treasury par curve: all published tenors



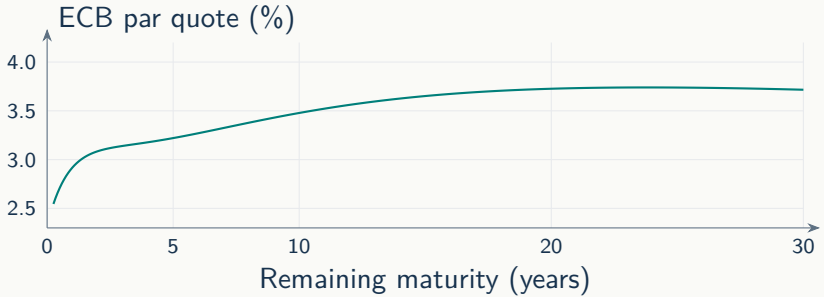
15 October 2026. All 14 published tenors, including the bill end. Treasury quotation conventions.

The short end deserves a closer look



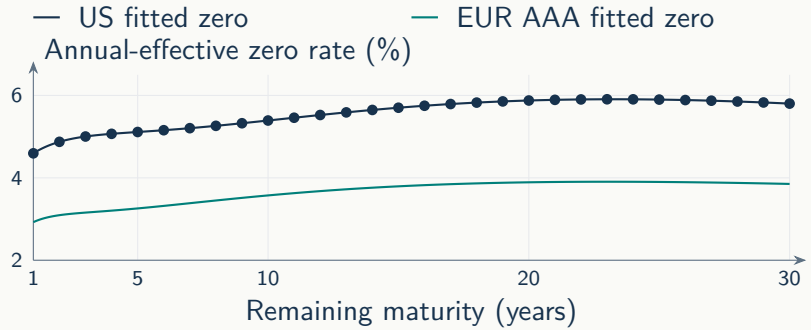
Same 15 October observations. A zoom changes visibility, not the data.

Euro-area AAA: 358 fitted maturities



14 October 2026. Published monthly grid, 3 months to 30 years; fitted points are not independent quotes.

Fitted zero curves on a common date



9 October 2026. Continuous source yields converted by $i = \exp(z) - 1$. Different currencies and samples.

Spot, par and YTM are different objects

Rate	Pricing question
Spot $i_t(T)$	What discounts one payment at T?
Par coupon rate	Which coupon makes a new bond worth F?
Bond-specific i^{YTM}	Which single rate fits this coupon schedule's price?

More plotted maturities make the fitted shape visible. They do not add independent market quotes.

A non-flat curve discounts dates differently

$$p_t = \frac{C}{1 + i_t(1)} + \frac{C + F}{[1 + i_t(2)]^2}.$$

Use each payment's own spot rate. The coupon rate does not choose the discount rate.

Bootstrapping a two-year discount factor

$$d_t(1) = 0.96, \quad C = 5, \quad F = 100, \quad p_t = 100.$$

$$100 = 5(0.96) + 105 d_t(2) \quad \Rightarrow \quad d_t(2) = 0.906667.$$

$$i_t(2) = d_t(2)^{-1/2} - 1 = 5.021\%.$$

A forward loan is agreed today



The future loan rate is fixed before the money changes hands.

Two ways to lock a two-year payoff

Route A	Route B
Buy a two-year zero.	Invest for one year.
Lock the whole two-year return today.	Lock year 2 using a forward agreed today.

Same initial investment, final date, currency, credit and collateral assumptions.

Exact no-arbitrage across the two routes

$$\boxed{[1 + i_t(2)]^2 = [1 + i_t(1)][1 + f_t(1, 2)].}$$

The forward rate adjusts until the locked final payoffs match.

Worked forward rate

$$i_t(1) = 4\%, \qquad i_t(2) = 5\%.$$

$$f_t(1,2) = \frac{1.05^2}{1.04} - 1 = 6.0096\%.$$

$$100(1.05)^2 = 100(1.04)(1.060096) = 110.25.$$

Replicating the forward with two zeros

Position	Today	Year 1	Year 2
Short 100-face one-year zero	$+96.1538$	-100	0
Buy 106.0096-face two-year zero	-96.1538	0	$+106.0096$
Net	0	-100	$+106.0096$

A future loan is a portfolio of dated claims. F is face value; f is the forward rate.

A mispriced forward creates a locked profit

Suppose an otherwise identical quoted forward lets you lend at 8%.

Borrow synthetically at 6.0096%, then lend the same 100 at 8%.

$$\text{Year-2 net cash flow} = 108 - 106.0096 = 1.9904.$$

A forward is a price; a forecast is a belief

Known today	Still uncertain
Locked $f_t(1,2) = 6.0096\%$	The one-year spot rate observed next year.
Its value follows from today's discount factors.	Its expectation needs a model or survey.

No-arbitrage links traded prices. It does not set the physical probability of future outcomes.

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Expectations and term premia

How much of a long yield can we interpret as a forecast?

Rolling over leaves the second rate unknown

Year-2 rate after a 4% first year	Final wealth from 100
2%	$100(1.04)(1.02) = 106.08$
8%	$100(1.04)(1.08) = 112.32$
Two-year zero locked at 5%	110.25

An unhedged roll-over is risky. The forward-locked strategy on earlier slides is different.

Compounding averages forward rates

$$[1 + i_t(n)]^n = \prod_{k=0}^{n-1} [1 + f_t(k, k+1)].$$

$$\ln[1 + i_t(n)] = \frac{1}{n} \sum_{k=0}^{n-1} \ln[1 + f_t(k, k+1)].$$

$f_t(0, 1) = i_t(1)$. These are rates locked today, not expected future spot rates.

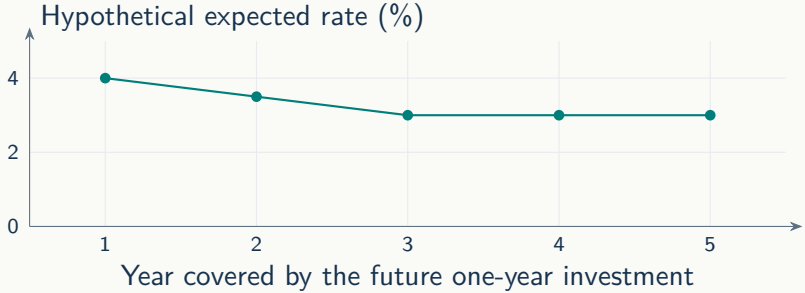
The expectations hypothesis adds an assumption

$$f_t(k, k+1) \approx \mathbb{E}_t \, i^{(1)}_{t+k}.$$

It treats the forward as an estimate of the future one-year rate, with no forward risk premium.

$i^{(1)}_{t+k}$: the one-year market rate prevailing at that future date.

An explicit expected short-rate path



$$\text{Arithmetic average} = \frac{4 + 3.5 + 3 + 3 + 3}{5} = 3.30\%.$$

Why a term premium can appear

A short-horizon holder of a long bond faces resale-price risk.

Investors may demand compensation for exposure to bad outcomes.

A bond that hedges the investor's bad states can instead command an insurance value.

Add the premium as a separate component

$$\underbrace{3.30\%}_{\text{expected-rate component}} + \underbrace{0.70\%}_{\text{term premium}} = \underbrace{4.00\%}_{\text{five-year yield}} .$$

The same expected path can support a different long yield if investors price the risk differently.

The yield decomposition, now defined

$$i_t(n) \approx \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_t i_{t+k}^{(1)} + TP_t(n).$$

Symbol	Meaning
n	Years to maturity
$\mathbb{E}_t i_{t+k}^{(1)}$	Expected one-year rate in year k
$TP_t(n)$	Term premium for that maturity

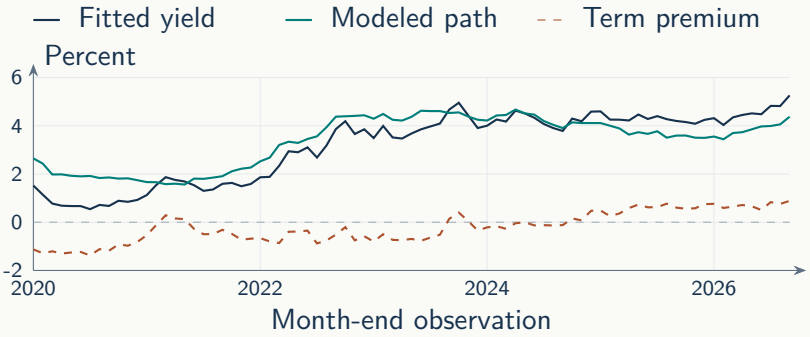
Approximation under a specified term-structure model.

One observed yield can have different stories

Story	Expected-rate component	Term premium	Yield
A	3.30%	0.70%	4.00%
B	3.80%	0.20%	4.00%

The yield alone cannot tell us which component investors changed.

An estimated US decomposition over time



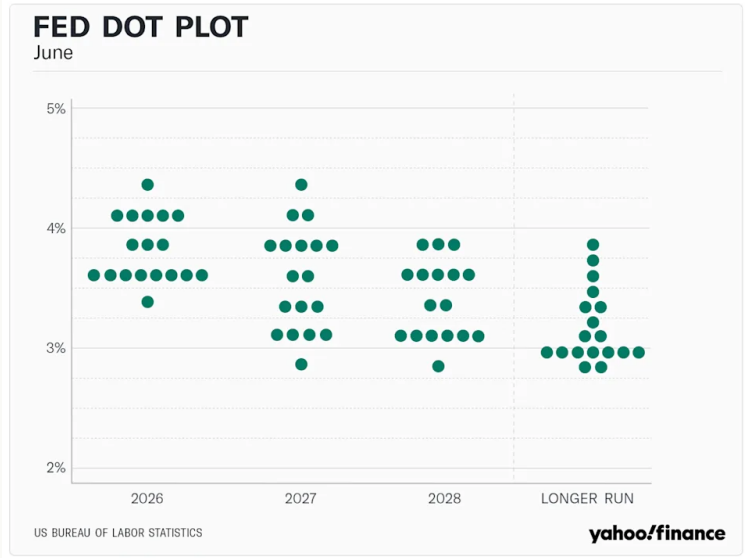
NY Fed ACM model, ten-year maturity. January 2020–September 2026. Components are estimates.

Reading the model snapshot

$$4.37\% + 0.89\% = 5.26\% \quad \text{approximately.}$$

30 September 2026	ACM estimate
Modeled rate-path component	4.3740%
Term premium	0.8850%
Fitted ten-year yield	5.2590%

The two components are model estimates. Their sum uses unrounded data.



Fed SEP, 17 June 2026. The supplied graphic's BLS credit is incorrect.

What each dot means

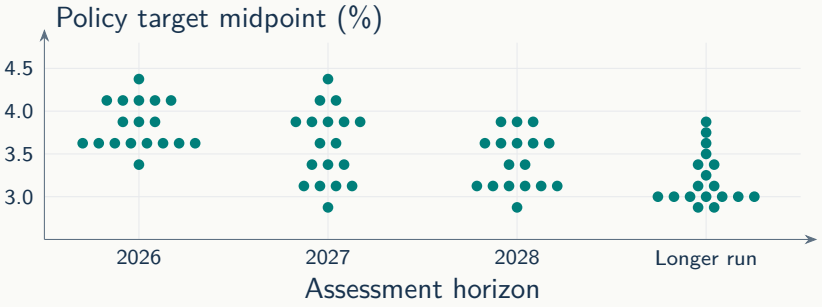
One official's assessment of appropriate policy at that year-end.

The assessment depends on that official's economic outlook.

Dots are anonymous. We cannot join a person's path across columns.

The dispersion is across officials. It is not a probability distribution of future policy outcomes.

June's individual assessments and medians



Official medians, rounded: 3.8%, 3.6%, 3.4%, 3.1%. Editable replot from the Fed's Figure 2.

Three objects to keep separate

Object	Whose information?	What is it?
Fed dots	Individual officials	Conditional policy assessments
Forward $f_t(T_1, T_2)$	Market prices today	Rate on a locked future interval
Expectation $\mathbb{E}_t i^{(1)}$	A model or survey	Belief about a future market rate

A cut today can coexist with rising long yields

Hypothetical experiment	Before	After
Current policy target $\bar{i}$	4.00%	3.75%
Expected average future short rates	3.30%	3.40%
Term premium	0.70%	0.90%
Long yield, approximately	4.00%	4.30%

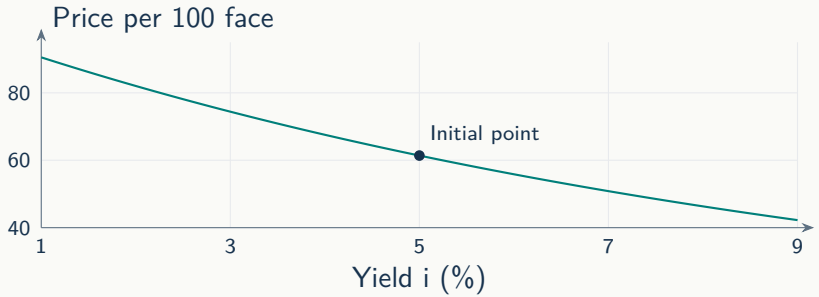
The current setting is only one part of the path. This is an invented scenario.

LECTURE 3

Duration and convexity

How does a discount-rate move change the price?

The price–yield relation is curved



Ten-year zero, fixed face value 100. In this section i is the yield varied in the pricing formula.

Modified duration is the normalized slope

$$D_{\text{mod}} = -\frac{1}{p}\frac{dp}{di}.$$

$$\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta i.$$

A 50 bp increase means $\Delta i = 0.005$. The negative sign gives the price loss.

A zero's duration follows by differentiation

$$p = F(1+i)^{-T}, \qquad \frac{dp}{di} = -TF(1+i)^{-T-1}.$$

$$\boxed{D_{\text{mod}} = \frac{T}{1+i}.}$$

$$T = 10, \quad i = 5\% \quad \Rightarrow \quad D_{\text{mod}} = 9.5238.$$

Coupon duration weights the payment dates

Date k	Cash flow	PV weight
1	4	3.92%
2	4	3.73%
3	104	92.35%

$$D_{\text{Mac}} = \sum_k k\, w_k = 2.8844 \text{ years.}$$

Macaulay and modified duration

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+i}.$$

$$D_{mMac} = 2.8844 \quad \Rightarrow \quad D_{\text{mod}} = 2.7470 \quad \text{at } i = 5\%.$$

Annual coupon convention. Payment timing gives Macaulay duration; the derivative gives modified duration.

DV01 converts sensitivity into money

$$DV01 \approx V D_{\text{mod}} \times 10^{-4}.$$

$$V = 1{,}000{,}000, \quad D_{\text{mod}} = 8 \quad \Rightarrow \quad DV01 = 800.$$

DV01 is currency units per basis point. One bp $= 0.01$ percentage point $= 0.0001$.

A ten-basis-point move

$$\Delta i = +10 \text{ bp} \quad \Rightarrow \quad \Delta V \approx -800 \times 10 = -8{,}000.$$
$$\Delta i = -10 \text{ bp} \quad \Rightarrow \quad \Delta V \approx +8{,}000.$$

The linear approximation is symmetric. The exact price curve is curved.

Duration comes from a Taylor expansion

$$p(i + \Delta i) = p(i) + p'(i)\Delta i + \frac{1}{2}p''(i)(\Delta i)^2 + O((\Delta i)^3).$$

The first derivative is the tangent. The second derivative measures curvature.

Cash flows and maturity are fixed while the yield parameter changes.

Convexity normalizes the second derivative

$$\mathcal{C} = \frac{1}{p}\frac{d^2 p}{di^2}.$$

$$\boxed{\frac{\Delta p}{p} \approx -D_{\text{mod}}\Delta i + \frac{1}{2}\mathcal{C}(\Delta i)^2.}$$

For a noncallable bond with fixed positive payments, the price–yield convexity is positive.

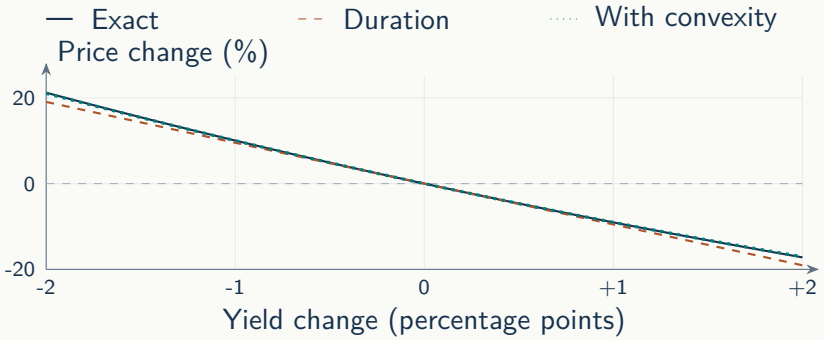
A zero's convexity is explicit

$$\frac{d^2p}{di^2} = T(T+1)F(1+i)^{-T-2}.$$

$$\boxed{\mathcal{C} = \frac{T(T+1)}{(1+i)^2}.}$$

$$T = 10, \quad i = 5\% \quad \Rightarrow \quad \mathcal{C} = 99.7732.$$

Exact, tangent and quadratic repricing



Ten-year zero, initial yield 5%, face 100. The quadratic follows the exact curve more closely locally.

Work out a one-percentage-point move

Yield move	Exact	Duration	With convexity
-1 point	+10.042%	+9.524%	+10.023%
+1 point	-9.043%	-9.524%	-9.025%

$$\tfrac{1}{2}\mathcal{C}(0.01)^2 = 0.0049887 \approx 0.499\%.$$

The asymmetry has economic meaning

For equal rate moves, the gain from a fall exceeds the loss from a rise.

More distant fixed payments generally amplify the curvature.

Positive convexity is valuable, but securities with different convexity need not have the same price or yield.

Sensitivity is conditional on the experiment

A single-yield bump isolates one price-risk channel.

Curve twists, credit spreads and changing cash flows need additional exposures.

Duration matching removes a local parallel-move exposure. It does not make a portfolio riskless.

A liability changes what counts as risky

Ten-year nominal liability of 100	Investment choice
Match the cash flow	A ten-year default-free zero pays 100 on the required date
Roll short bills	The future reinvestment path is uncertain
Trade a long bond over one year	The sale price is uncertain

A bond fund usually keeps exposure beyond one maturity

A single bond has a contractual maturity payment.

A rolling bond fund continually replaces holdings and retains a duration exposure.

The relevant comparison is the investor's cash-flow need and holding horizon.

LECTURE 3

Hedging and market cases

How does a finance view become a position with a measurable risk?

News lab: income horizon or price trade?

A WSJ bond-strategy column asks investors to identify their objective.

Compare a fixed spending need with a forecast of a bond-price rally.

Telis Demos, WSJ, 8 October 2026. Discussion prompt, not a trade recommendation.

News lab: buying during a Treasury sell-off

FT reporting describes investors adding Treasuries after a substantial rise in yields.

A higher starting yield improves the quoted entry rate; a further rate rise can still create a near-term loss.

Clarfelt and Duguid, Financial Times, 13 October 2026. Separate the report from our hypothetical exercise.

Two real auctions, two different questions

Auction	Offering	High yield	Bid-to-cover
7 Oct: ten-year note reopening	39 bn	5.300%	2.77
8 Oct: thirty-year bond reopening	22 bn	5.618%	2.54

TreasuryDirect results, October 2026. These are auction yields, not coupon rates or total returns.

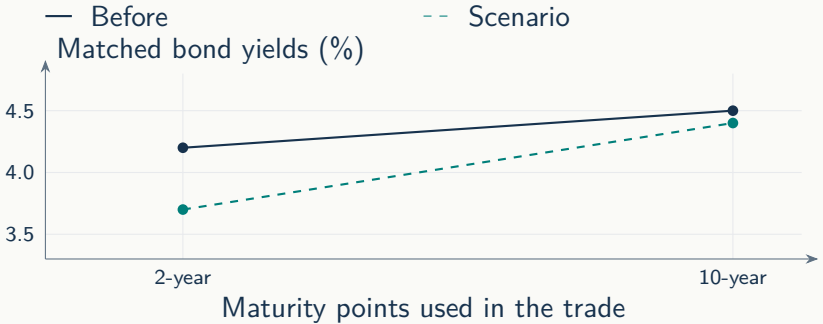
An auction tail needs a market benchmark

$$\text{Tail} = \text{auction stop yield} - \text{when-issued yield at the auction deadline.}$$

$$5.30\% - 5.28\% = 0.02 \text{ points} = 2 \text{ bp.}$$

Illustrative numbers. Same issue, timestamp and quote convention are required.

A curve view needs two maturity scenarios



Two risk points in an invented trade scenario. This is not a sparse empirical curve estimate.

This scenario is a bull steepening

$$\text{Slope} = i_{10} - i_2.$$

$$4.50\% - 4.20\% = 0.30\% \quad \longrightarrow \quad 4.40\% - 3.70\% = 0.70\%.$$

Yields fall. The long-minus-short yield spread widens.

Expressing the relative view

Leg	Benefit	Risk in this scenario
Buy two-year bond	Its yield falls 50 bp	Positive price contribution
Short ten-year bond	Its yield rises	Its 10 bp fall produces a loss

A slope view needs a risk hedge. Equal face amounts do not match rate sensitivity.

Match DV01, rather than market values

Position	Market value	Modified duration	DV01
Long two-year	1,000,000	1.8	180 / bp
Short ten-year	?	8.0	180 / bp

$$V_{\text{short}} = \frac{180}{8(0.0001)} = 225{,}000.$$

Work out the scenario profit and loss

Leg	Yield move	Approximate P&L
Long two-year	-50 bp	$+180(50) = +9{,}000$
Short ten-year	-10 bp	$-180(10) = -1{,}800$
Combined	40 bp steepening	$+7{,}200$

First-order repricing before carry, financing, trading costs and convexity.

A small parallel move cancels at first order

$$\Delta i_2 = \Delta i_{10} = +25 \text{ bp.}$$

$$\Delta V \approx -180(25) + 180(25) = 0.$$

The spread does not change. Higher-order risk and financing remain.

Counterfactual: the curve flattens

Leg	Alternative yield move	Approximate P&L
Long two-year	-10 bp	$+1{,}800$
Short ten-year	-50 bp	$-9{,}000$
Combined	40 bp flattening	$-7{,}200$

Both bonds rally, but this steepener loses.

A cut with a sell-off: the complete calculation

$$\Delta i_{10} \approx +0.10 + 0.20 = +0.30 \text{ percentage points.}$$

$$V = 1{,}000{,}000, \quad D_{\mathrm{mod}} = 8 \quad \Rightarrow \quad \Delta V \approx -24{,}000.$$

$$\mathcal{C} = 80 \quad \Rightarrow \quad \text{quadratic correction} = +360.$$

Questions before taking the trade

Which future cash flow, rate shock and holding horizon are being priced?

How does the forecast differ from the forward curve or market baseline?

What happens under the opposite curve move, after financing and costs?

LECTURE 3

Optional foundations

How do dates, states and compounding fit one pricing framework?

General forward intervals

$$[1 + f_t(T_1, T_2)]^{T_2 - T_1} = \frac{d_t(T_1)}{d_t(T_2)}.$$

$$f_t(T_1, T_2) = \left[\frac{d_t(T_1)}{d_t(T_2)}\right]^{1/(T_2 - T_1)} - 1.$$

Annual-effective forward convention. Actual contracts can use simple interest and specific day counts.

Continuous and annual-effective spot rates

$$d_t(T) = e^{-z_t(T)T} = [1 + i_t(T)]^{-T}.$$

$$i_t(T) = e^{z_t(T)} - 1.$$

The Fed and ECB zero inputs use continuous rates z. Our matched-date comparison converts them to i.

The par coupon comes from discount factors

$$F = c_{\mathrm{par}} F \sum_{k=1}^{n} d_t(k) + F d_t(n).$$

$$\boxed{c_{\mathrm{par}} = \frac{1 - d_t(n)}{\sum_{k=1}^{n} d_t(k)}.}$$

Annual coupon dates and unit year fractions in this simplified formula.

Two states: equal average payoff, different price

Payoff	Good state	Bad state	Expected payoff	Price
A	100	0	50	40.00
B	0	100	50	56.15

Original two-state example. Both states have probability 1/2. Payments in the bad state are more valuable.

A stochastic discount factor prices states

$$p_t = \mathbb{E}_t[m_{t+1} X_{t+1}].$$

$$p_t = \mathbb{E}_t[m]\,\mathbb{E}_t[X] + \mathrm{Cov}_t(m, X).$$

X: future payoff. m: value today per unit paid in that state, adjusted for its probability.

Optional liability hedge: match value and duration

Liability: 100 in ten years, $i = 5\%$.

$$w_1 + w_{20} = 1, \qquad 1w_1 + 20w_{20} = 10.$$

$$w_1 = \frac{10}{19}, \qquad w_{20} = \frac{9}{19}.$$

Weights are current market-value shares in one- and twenty-year zeros.

LECTURE 3

Reading

Where can we deepen the finance and market perspective?

Reading and next questions

Bodie, Kane and Marcus: pricing, term structure and bond-risk management.

The two supplied finance courses: replication, realized returns, auctions and immunization.

Fed, Treasury and ECB sources; WSJ and FT discussion readings via MIT.

Exact chapters, page mappings, dates and links are in the speaker notes and reading list.